

The Annual Training Periodization of 8 World Champions in Orienteering

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One year of training data from 8 elite orienteers were divided into a transition phase (TP), general preparatory phase (GPP), specific preparatory phase (SPP), and competition phase (CP). Average weekly training volume and frequency, hours at different intensities (zones 1–3), cross-training, running, orienteering, interval training, continuous training, and competition were calculated. Training volume was higher in GPP than TP, SPP, and CP (14.9 vs 9.7, 11.5, and 10.6 h/wk, $P < .05$). Training frequency was higher in GPP than TP (10 vs 7.5 sessions/wk, $P < .05$). Zone 1 training was higher in GPP than TP, SPP, and CP (11.3 vs 7.1, 8.3, and 7.7 h/wk, $P < .05$). Zone 3 training was higher in SPP and CP than in TP and GPP (0.9 and 1.1 vs 1.6 and 1.5 h/wk, $P < .05$). Cross-training was higher in GPP than SPP and CP (4.3 vs 0.8 h/wk, $P < .05$). Interval training was higher in GPP than TP, SPP, and CP (0.7 vs 0.3 h/wk, $P < .05$). High-intensity continuous training was higher in GPP than CP (0.9 vs 0.4 h/wk, $P < .05$), while competition was higher in SPP and CP than in TP and GPP (1.3 and 1.5 vs 0.6 and 0.3 h/wk, $P < .01$). In conclusion, these champion endurance athletes achieved a progressive reduction in total training volume from GPP to CP via a shortening of each individual session while the number of training sessions remained unchanged. This decrease in training volume was primarily due to a reduction in the number of hours of low-intensity, non-sport-specific cross-training.

Keywords: elite athletes, intensity distribution, endurance

The principle of training “periodization” was first introduced by Matveyev in the 1960s,¹ based on analyses of successful Soviet athletes from a variety of sports during the 1950s and '60s. A key feature of the Matveyev model was early emphasis on high training volume and a transition to higher training intensity (with reduced volume) as competition periods approached. A second feature of the model was increasing training specificity throughout the annual cycle. Since then, a number of periodization models and training philosophies have emerged.^{2,3} However, data presenting quantitative details regarding training periodization in practice or its effect on performance remain remarkably scarce 50 years later.⁴

Training periodization has been described as the purposeful sequencing of different training units (long-duration, medium-duration, and short-term training cycles and sessions) so that athletes can attain the desired state and planned results.⁵ Over recent decades, the degree of professionalism among coaches and athletes in endurance sports has increased. This has, among other things, resulted in a substantial increase in training volume. At elite senior level in endurance sports today, training volumes of up to 1000 h/y are reported.⁶ Studies of cycling,⁷ rowing,^{8,9} running,^{10,11}

and cross-country skiing^{12,13} show that 80% to 90% of this training is performed at an intensity below the first lactate turn point. The remaining 10% to 20%, constituting 100 to 150 yearly training sessions, is performed above the first lactate turn point (high-intensity interval training [HIT]).^{6,12,13} Studies that have explored the effects of HIT in elite athletes have primarily been short-term comparisons of interval-training models using variations in duration, intensity, and recovery^{14,15} rather than how this training is periodized and executed across a typical training year. Despite the popularity of periodization concepts and

terminology, it remains unclear how elite endurance athletes actually execute training plans and integrate HIT across the annual cycle.

In endurance sports, many successful athletes now supplement sport-specific endurance training with alternative activity forms such as running, cycling, swimming, and cross-country skiing.^{16–19} This practice is often termed cross-training. Arguments supporting the inclusion of nonspecific training include injury prevention, general central capacity effects, strengthening weak links, and avoidance of training monotony.^{17,20} However, few data are available specifically connecting cross-training practice to performance. Based on our personal experience, the volume of nonspecific training incorporated into the annual training of elite athletes varies much more than specific training volume, but this is not well documented.

The aim of this study was to describe and analyze changes across a full year of training among 8 world-champion endurance athletes from the same sport. We hypothesized that the training characteristics of these athletes would differ significantly across the transition phase, general preparatory phase, specific preparatory phase, and competition phase in terms of training duration, intensity, and frequency. We further hypothesized that the mode of delivery of HIT stimuli would vary in the different training phases, becoming more competition specific.

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Materials and Methods

Subjects

Eight (6 male and 2 female) former and current Norwegian elite orienteers were included in the study. The inclusion criterion was having won at least 1 individual world-championship gold medal during their career. In addition, all included athletes had systematically and comprehensively recorded their day-to-day training from junior through to senior level, and specifically during the year leading up to their most successful competition at senior level. Informed consent was obtained from all athletes to include their personal training data in analyses. Initially, training data for 10 world-champion orienteers were collated. Of these, 2 were excluded due to insufficient training documentation after inspection of their training diaries. Sylta et al^{21,22} recently reported high reliability of self-report training data from elite endurance athletes, in terms of both duration and intensity self-report accuracy. Table 1 describes the characteristics of these athletes.

Training Diaries and Systemization of Data

Athletes included in the investigation recorded their day-to-day training in training diaries designed by the Norwegian Orienteering Association.²³ In these diaries, athletes recorded time spent on different forms of training for each training session, along with the intensity of all aerobic endurance training. In addition, athletes described the session structure in a comments box. For example, interval length, number of repetitions, recoveries, heart rate, and

subjective feelings of exertion were recorded, as well as other comments related to the quality of the session.

Endurance training was classified into 3 intensity zones anchored around the lactate turn-point maximal lactate steady state as previously described.²⁴ Table 2 shows the intensity scale that the athletes used as a basis for categorizing their aerobic endurance training.

All training sessions completed during the most successful season of each athlete's career were systematized in Microsoft Excel (Microsoft Corp, Redmond, WA) and used in data analysis. Data were analyzed in relation to total training duration and frequency; the training time registered by the athlete for different activity forms—that is, running, cross-training (cross-country skiing, cycling, swimming) and orienteering; and time in each of the 3 intensity zones. Cross-training was defined as any endurance training that was not performed as running with or without a map. Training recorded as performed at intensity above the first lactate turn point (zones 2 and 3) was further categorized as HIT, continuous training, or competition. One athlete had not recorded the proportion of run training conducted with or without a map (running vs orienteering), so only the remaining 7 athletes were included in analyses of these 2 variables.

Definition of Phases

For analyses, the training year was divided into 4 training phases: transition phase (TP; October–November), general preparatory phase (GPP; December–March), specific preparatory phase (SPP; April–June), and competition phase (CP; July–September). These 4 phases were defined based on basic periodization models in the literature,^{1,5} the training data, and personal communications with the coaches and athletes (Figure 1).

Statistical Analyses

All training data were found to be normally distributed and are therefore presented as means with standard deviations unless otherwise stated. Statistical analyses were performed using IBM SPSS Statistics 19 (SPSS Inc, Chicago, IL). Data were initially calculated as mean training duration and frequency per month. Total training time and number of sessions for each phase were then calculated. This value was subsequently divided by the number of days in that phase and multiplied by 7 to give an average weekly training duration/frequency for each phase. This process was repeated to determine average weekly training hours at different intensities (zone 1, zone 2, and zone 3); average time spent cross-training, running without a map, and running with a map (orienteering);

Table 1 Demographics of Athletes Included in Analyses, Median (Range)

Men (n = 6)

Women (n = 2)

World championship gold medals (career total) 3 (1–7) 3 (2–4)

Age at time of analyzed season (y) 25 (21–31) 29 (25–32)

Height (m) 1.81 (1.77–1.85) 1.73 (1.68–1.78)

Weight (kg) 71 (56–78) 56 (55–57)

Maximal oxygen uptake ($\text{mL} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) 83 (81–85) 72 (70–75)

Annual training volume (h) 636 (547–731) 613 (537–689)

Table 2 The Intensity Zones Used for Categorization of Aerobic Training

Intensity zone % of HRmax Blood lactate (mmol/L)^a Examples of training models

3 88–100 4.0–10.0 High-intensity continuous exercise or interval training with a high level of exertion and recovery time equal to 50–90% of the work interval time.

2 83–87 2.5–4.0 Natural interval training, intensive continuous training, or long intervals with recoveries equal to 20–30% of the work interval time.

1 55–82 < 2.5 Moderate-intensity continuous training, recovery sessions, and low-intensity continuous training.

Abbreviations: HRmax, maximal heart rate.

a Normative blood lactate concentration values based on red-cell lysed blood.

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Figure 1 — Training phases and monthly training time for strength training and endurance training in each of the 3 intensity zones (bars) and training frequency (line) in the year leading up to winning a world championship. Values are group means.

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and what proportion of high-intensity training (zones 2 and 3) was conducted as interval training, continuous training, or competition. To identify whether there were statistically significant differences in these training variables between the 4 phases, a 1-way repeated-measures analysis of variance was performed to identify significant differences in training characteristics across the 4 training phases. The Bonferroni correction was applied to the α level ($P < .05$) to guard against inflated type I error risk with multiple comparisons.

Results

In total, 3837 training sessions were systematized in the current analysis. Athletes reported 480 ± 65 training sessions in the season in which they won a world championship.

Figure 1 presents a summary overview of training volume, frequency, and intensity distribution for the 8 subjects over the entire year of their best performance. The TP from the end of the previous season to the beginning of the new training cycle showed considerable variation in training volume and frequency among the 8 athletes, ranging from 6 to 14 hours and 4 to 10 sessions/wk. However, the move to the GPP was consistently characterized by higher total training volume (~50% mean increase) than in the preceding TP that separated 2 training annual cycles ($P = .001$).

Average total training time per week was significantly higher in GPP than TP ($P = .001$), SPP ($P = .012$), and CP ($P < .001$; Figure 2). Similarly, weekly training time in zone 1 was significantly higher in GPP than TP ($P = .006$), SPP ($P = .010$), and CP ($P < .001$; Figure 3).

Average number of training sessions per week was significantly higher in GPP than in TP ($P = .011$), while there were no differences in average number of training sessions per week among the other training phases (Table 3). Weekly training time in zone 3 was significantly higher in SPP and CP than in either TP ($P < .05$) or GPP ($P < .05$), while there were no significant differences between any of the training phases with regard to time in zone 2 (Table 3). Average hours per week of non-sport-specific cross-training were significantly higher in GPP than SPP ($P = .025$) and CP ($P = .048$; Figure 4), with no difference between the other training phases. There were no significant differences in hours per week spent on general running training (without a map) or orienteering between any of the training phases. Average hours per week spent on HIT were significantly higher in GPP than TP ($P = .013$), SPP ($P = .018$), and CP ($P = .045$; Figure 5). Average hours per week of high-intensity continuous training were significantly higher in GPP than CP ($P = .039$).

Average weekly hours of competition were significantly higher in both SPP and CP than in TP ($P = .003$ and $P < .001$, respectively) and GPP (both $P < .001$).

Discussion

The main finding of the study was that the annual training pattern of 8 world-champion orienteers consistently showed a significant reduction in training volume from GPP to SPP and CP. This was achieved through reduction in nonspecific training loads (cross-training),

while specific running training volume was quite stable across GPP, TP, and CP. From GPP and throughout the season, training frequency remained quite stable at 9 to 10 sessions/wk. Figure 2 — Weekly training volume (left panel) and frequency (right panel). Values are group mean (bold line) and individual data for each athlete.

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Figure 3 — Weekly training time in intensity zones 1, 2, and 3. Values are group mean (bold line) and individual data for each athlete. *Mean significantly different from transition phase ($P < .05$). ‡Mean significantly different from specific preparatory phase ($P < .01$). #Mean significantly different from competition phase ($P < .01$). †Mean significantly different from general preparatory phase ($P < .05$).

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Table 3 Differences in Training Variables Across the 4 Phases, Mean $\pm$ SD

Transition Phase General Preparatory Phase Specific Preparatory Phase Competition Phase
Annual Total

Absolute (/wk) % of

total AT Absolute (/wk) % of

total AT Absolute (/wk) % of

total AT Absolute (/wk) % of

total AT Absolute % of

total AT

Weekly training durationa

9.7 $\pm$ 3.4 h — 14.9 $\pm$ 1.6 h*†‡ — 11.5 $\pm$ 1.5 h — 10.6 $\pm$ 1.8 h — 629 $\pm$ 75 h —

Weekly training sessions

8 $\pm$ 3 — 10 $\pm$ 2 * — 10 $\pm$ 1 — 9 $\pm$ 1 — 480 $\pm$ 65 —

Zone 1 7.1 $\pm$ 2.9 h 81 $\pm$ 7 11.3 $\pm$ 1.8 h*†‡ 84 $\pm$ 6 8.3 $\pm$ 1.5 h 77 $\pm$ 6 7.7 $\pm$ 2.0 h 77 $\pm$ 6 464 $\pm$ 83 h 80 $\pm$ 5

Zone 2 0.7 $\pm$ 0.3 h 9 $\pm$ 4 1.0 $\pm$ 0.5 h 8 $\pm$ 4 0.8 $\pm$ 0.4 h 8 $\pm$ 4 0.7 $\pm$ 0.4 h 7 $\pm$ 4 44 $\pm$ 17 h 8 $\pm$ 3

Zone 3 0.9 $\pm$ 0.3 h 10 $\pm$ 5 1.0 $\pm$ 0.4 h 8 $\pm$ 3 1.5 $\pm$ 0.4 h*# 15 $\pm$ 4 1.5 $\pm$ 0.4 h*# 16 $\pm$ 5 67 $\pm$ 15 h 12 $\pm$ 3

Running, orienteering specific (n = 7)

2.7 $\pm$ 1.9 h 27 $\pm$ 14 3.2 $\pm$ 1.4 h 24 $\pm$ 9 4.3 $\pm$ 0.5 h 41 $\pm$ 4 3.2 $\pm$ 0.7 h 33 $\pm$ 6 177 $\pm$ 41 h 30 $\pm$ 5

Running, general (n = 7)

5.0 $\pm$ 1.7 h 57 $\pm$ 20 6.2 $\pm$ 2.0 h 45 $\pm$ 11 5.4 $\pm$ 0.7 h 51 $\pm$ 5 5.7 $\pm$ 1.7 h 58 $\pm$ 7 296 $\pm$ 73 h 51 $\pm$ 7

Cross-training 1.5 $\pm$ 1.2 h 17 $\pm$ 12 4.3 $\pm$ 1.7 h*†‡ 32 $\pm$ 14 0.8 $\pm$ 0.7 h 8 $\pm$ 6 0.8 $\pm$ 1.0 h 8 $\pm$ 10 109 $\pm$ 27 h 19 $\pm$ 6

Interval trainingb 0.3 $\pm$ 0.1 h 21 $\pm$ 7 0.7 $\pm$ 0.2 h*†‡ 37 $\pm$ 14 0.3 $\pm$ 0.1 h 15 $\pm$ 5 0.3 $\pm$ 0.2 h 15 $\pm$ 7 23 $\pm$ 6 h 22 $\pm$ 6

High-intensity continuous trainingb

0.7 $\pm$ 0.3 h 42 $\pm$ 19 0.9 $\pm$ 0.5 h‡ 44 $\pm$ 18 0.6 $\pm$ 0.5 h 24 $\pm$ 18 0.4 $\pm$ 0.3 h 17 $\pm$ 12 33 $\pm$ 20 h 31 $\pm$ 14

Competitionb 0.6 $\pm$ 0.3 h 37 $\pm$ 20 0.4 $\pm$ 0.1 h 19 $\pm$ 10 1.4 $\pm$ 0.4 h*# 61 $\pm$ 14 1.5 $\pm$ 0.3 h*# 68 $\pm$ 8 48 $\pm$ 11 h 47 $\pm$ 11

Abbreviations: AT, aerobic training.

Note: Absolute values represent average number of training hours per week at different intensities and in different training modes. Values are also presented as a percentage of total aerobic training in each phase.

a Includes strength, power, and agility training. b Percentage values for interval training, continuous training, and competition are fractions of high-intensity training (zones 2 and 3) rather than total training.

*Significant difference from transition phase, $P < .05$. #Significant difference from general preparatory phase, $P < .05$. †Significant difference from specific preparatory phase, $P < .05$. ‡Significant difference from competition phase, $P < .05$.

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A second key finding was that the reduction in total training volume occurs primarily due to a reduction in the number of training hours in intensity zone 1. Although training time in zone 3 is increased in SPP and CP, total training time above the first lactate turn point (zones 2 + 3) was consistent across these phases.

Training Volume and Frequency Periodization

All 8 athletes reported increased weekly training volume in GPP compared with SPP and CP. A common feature among all 8 athletes is a gradual reduction in training volume from one month to the next during SPP (Figure 1). In isolation this volume reduction is in agreement with “best practice” training philosophies presented in the literature.^{1–3} However, the major source of this decrease is reduced nonspecific training, not specific training. Running loads remained relatively stable at ~9 hours effective training per week throughout GPP, SPP, and CP (Figure 4). It is also apparent that, other than a small increase from TP to GPP, training frequency appears to be maintained throughout the training year. Hence, the observed reduction in training volume is primarily achieved by a shortening of each training session. Our observations are in accordance with previous research showing that elite athletes reduce training volume to obtain optimal effects from a tapering period and reach peak performance at the desired time.^{25,26} One difference between the observation in the current study and previous reported recommendations from tapering studies is the time frame for reduction in duration and training load. Figure 1 shows that athletes typically increase their training volume during the first month of CP and then reduce it the last 6 to 8 weeks

before desired peak performance. However, the reduction in training volume in this period appears to be typically not more than ~20%. A large meta-analysis of 30 studies on elite athletes presented by Bosquet et al²⁶ found that a reduction in training volume of 40% to 60% over the last 14 days results in the greatest performance improvement. In comparison, the observed reduction in training volume in the current study is rather modest. However, it is worth noting that the current study looked at average training over a month or phase, and we cannot exclude the possibility that shorter transient changes in training volume over the final 1 to 2 weeks before the world championships may have been present. Figure 1 also shows that both training volume and frequency are significantly higher during GPP than during TP. The progressive increase in training volume and frequency (Figures 1 and 2) is more or less identical for all the athletes in the study and, to an extent, consistent with traditional training philosophies^{1–3} and advice regarding training progression presented in the literature.²⁷ It is reasonable to assume that the purpose of such a progression is to prevent injury or illness.

Intensity Distribution

Athletes in the current study performed significantly more zone 3 training during SPP and CP than during TP and GPP. During these 2 phases athletes trained on average 1.5 h/wk at this intensity level—almost twice as much as during TP and GPP. This progression in high-intensity training is as expected and to an extent consistent with conventional coaching philosophies.³ The proportion of total high-intensity training (zones 2 and 3) is relatively similar for all of the athletes, with training in zones 2 and 3 comprising 19%,

Figure 4 — Weekly hours of running and cross-training during each of the 4 phases. Values are group mean (bold line) and individual data for each athlete. *Mean significantly different from specific preparatory phase ($P < .05$). †Mean significantly different from competition phase ($P < .05$).

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Figure 5 — Weekly hours of high-intensity interval training, high-intensity continuous training, and competition during each of the 4 phases. Values are group mean (bold line) and individual data for each athlete. *Mean significantly different from transition phase, $P < .05$. †Mean significantly different from general preparatory phase, $P < .05$. #Mean significantly different from specific preparatory phase, $P < .05$. ‡ Mean significantly different from competition phase, $P < .05$.

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16%, 23%, and 23% of total aerobic training, respectively, during TP, GPP, SPP, and CP. This distribution of high-intensity training is in accordance with previous studies on elite athletes in cycling,⁷ running,¹¹ rowing,⁹ and cross-country skiing.²⁸ However, Seiler and Tønnessen⁶ reported that elite endurance athletes spent somewhat less—only ~10%—of their total training time at high intensity.

During GPP, training duration is high, with a lower fraction of training conducted at high intensity. However, even during TP and GPP, athletes typically trained 2 or 3 high-intensity sessions per week. Training some sessions at high intensity throughout the annual cycle agrees well with the recommendations of Tschene,²⁹ who suggested that maintaining intensive and specific training throughout the training year allows athletes to optimize the development of both endurance and technique across the annual cycle. However, too great an emphasis on intensive and specific training early in the season can lead to overtraining or premature peaking.^{20,30}

Interval Training, Continuous Training, and Competition

In the research literature, high-intensity training and high-intensity interval training are terms often used interchangeably. However, the results of the current study show that these world-champion orienteers conducted the high-intensity portion of their training not only as interval training but also as continuous training and competition. The use of interval training was significantly higher in GPP than in the other 3 phases. In TP, GPP, SPP, and CP, respectively, 21%, 37%, 15%, and 15% of all high-intensity training was performed as interval training. Conversely, the use of competition was significantly higher in SPP and CP, with as much as 61% and 68% of total high-intensity training comprising competitive activity during these 2 phases. Competitive participation during these phases appears to be an essential component of training in order for athletes to reach their peak form. Competitions provide a holistic stimulus of physical, technical, tactical, and mental factors required for peak performance. Such a stimulus is difficult to achieve via high-intensity exercise training. The transition from high-intensity continuous and interval training to competition leads to increased specific load and may partly explain why the overall proportion of high-intensity training does not increase in these periods. It is interesting to note that all 8 of the athletes participated in competitions throughout the training year, although to a much lesser extent during TP and GPP. Further research is warranted to examine the differential effects of the same intensity and total duration of interval training, continuous training, and competition on physical capacity and performance.

Specific and Nonspecific Endurance Training

Some previous studies suggest that, in elite endurance athletes, $\text{VO}_{2\text{max}}$ may not be significantly improved by the use of cross-training.^{16,31} However, all 8 of the athletes in the

current study used alternative activity forms such as cross-country skiing, cycling, and swimming to supplement their sport-specific training. Cross-training made up, on average, 20% of all aerobic training across the training year. The number of hours cross-training was significantly higher in GPP than in SPP and CP. It is interesting to note that the larger total training volume in GPP is due to increased cross-training volume in this period. The aim of such training during GPP is to increase trainability and improve general aerobic capacity,^{19,27} as well as serving to reduce training monotony, which has been linked to negative outcomes.²⁰ The form of cross-training used by the athletes during this phase was primarily cross-country skiing. This

be a crucial criterion for cross-training to have a positive effect.^{16,32} Alternative training also appears important in creating variety and motivation. However, for all of the athletes, training became more sport-specific during SPP and CP, with almost all training during these phases conducted as running, with or without a map. During SPP and CP, on average only 8% of aerobic training was cross-training, compared with 32% during GPP. During SPP and CP, alternative activity forms appear primarily to have been used only as recovery training or during periods of injury.

It is not clear whether the observed organization was a result of deliberate long-term planning or merely self-organization in response to available energy and perceived needs. The progression and variation in the athletes' training across the annual cycle bears resemblance to traditional periodization models. However, similar to the block model,⁵ some types of training appear to have been emphasized more during certain periods (eg, strength training was prioritized more during GPP). The training of all of these athletes shows consistent variations across a number of variables.

Although such theoretical models provide a valuable framework for coaches and athletes, it is important that training be adapted in line with the sport's distinctive character, as well as the competition program, training conditions, and individual circumstance. The current data suggest that the training of elite endurance athletes is modified across the annual cycle via manipulation of a number of training-load variables including training duration, intensity, and frequency; activity form (specific vs nonspecific); and the mode of delivery of high-intensity training (interval vs continuous) to optimize performance in the world championships. We feel that the definition of training periodization presented in Issurin⁵ perhaps does not fully capture this. We therefore propose that periodization could rather be defined as the systematic and appropriate organization and dosage of training-load variables (training intensity, duration, frequency, and mode), both in the short and long term, to achieve peak performance at a specific time or at specific junctures in the competitive season.

Practical Applications

Despite some small discrepancies, the training periodization implemented by these 8 world champions in the most successful year of their athletic career is very similar. This is interesting considering that the training data collected span over 40 years, with the athletes included in the study between them collecting 25 world-championship gold medals from 1979 to 2012. This suggests that training culture in orienteering has not markedly changed over the last 4 decades.

Although further longitudinal studies on elite athletes from different nations and different sports are warranted, the current detailed training analysis of 8 world-champion endurance athletes leads to 3 specific recommendations for coaches and athletes:

- The mode of delivery of high-intensity training appears to be an important variable that can be manipulated to allow athletes to reach peak form at the desired time.

Interval training, high-intensity continuous training, and competition are all important components of the training process. However, during the SPP and CP, a large proportion of all high-intensity exercise should be conducted as competitions, where these provide a holistic stimulus of physical, technical, tactical, and mental factors required for peak performance.

- A steady progression in training volume from the TP through the GPP is recommended to ensure that performance is

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- To reach peak performance, it is recommended that non-sport-specific training be essentially eliminated during the SPP. Reductions in low-intensity training volume can be achieved by shortening training sessions modestly, without reducing training frequency.

Conclusion

These 8 world-champion orienteers achieved a progressive reduction in total training volume from GPP to CP via a shortening of each individual session, while the number of training sessions remained unchanged. This decrease in training volume was primarily due to a reduction in the number of hours of low-intensity exercise and reduced non-sport-specific cross-training. During the competitive season, only a modest proportion (22%) of all high-intensity training was performed as interval training. The remainder comprised high-intensity continuous exercise and competitive activity.

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